

HADI AMIRI

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SUMMARY

Applied AI Engineer with over 4 years of experience specializing in applied AI, RAG systems, and agentic workflows. Proven ability to deliver production generative AI solutions, including agentic AI platforms, hybrid retrieval pipelines, LLM evaluation, and PII de-identification services deployed on Azure and Kubernetes. Also brings strong cloud data engineering expertise: migrating complex ETL processes to Microsoft Fabric and Azure Databricks, optimizing data pipelines, and implementing robust data governance, leveraging data architecture and AI integration to drive innovation and efficiency.

WORK EXPERIENCE

Revolution Data Platforms | Applied AI Engineer

2024 – Present

/ Cloud Data Engineer / Data Platform Engineer

- Accelerated document ingestion time by 2.3x and reduced average RAG agent response time by approximately 80% (from 36s to 6.6s) on an air-gapped, Kubernetes-hosted agentic AI platform, while implementing hybrid dense+sparse retrieval on Qdrant with content-hash deduplication and a synthetic golden-dataset evaluation pipeline.
- Built a LangGraph agent toolbox featuring parallel validation and judge nodes, ensuring deterministic AI usage and human-gated workflows within the client's Azure AI Foundry tenant for data sovereignty, and developed a provider-agnostic client capable of switching between Anthropic, Azure OpenAI, and Azure AI Foundry without code modifications.
- Deployed a production PII de-identification service (Microsoft Presidio + RoBERTa NER) as an Azure ML managed endpoint, processing approximately 10 million clinical records with 94.9% redaction accuracy, and integrated it into the client's SSIS ETL via a thread-safe C# task supporting bulk loading.
- Transformed Excel-based financial reporting into a Microsoft Fabric lakehouse, processing over 1M records annually across 30+ tables sourced from a PeopleSoft ERP; implemented bronze/silver/gold Delta layers, event-driven Data Factory ingestion, parameterized PySpark transformation notebooks, data masking, and role-based access, thereby automating 7-8 hour manual tasks with scheduled workflows.
- Spearheaded an SSIS-to-cloud migration program for a federal government department, converting legacy ETL packages into Microsoft Fabric pipelines and Spark SQL notebooks over a landing/staging/mart lakehouse, incorporating ARM-template deployment, a 3-environment Dev/Staging/Prod workspace model, and Azure DevOps CI/CD.
- Engineered a behavior-based routing engine that classified 1,761 SSIS packages into 5 migration lanes by structural behavior, achieving 75.1% automated/template coverage, alongside a T-SQL to Spark SQL conversion pipeline that successfully processed 420 out of 420 production views with full fidelity.
- Delivered 24 clinical data extract packages end-to-end with confirmed live Fabric imports, featuring a 4-layer validation framework (structure, coverage, flow sanity, live import) and delivering a sign-off-quality architecture document accompanied by a manager-facing presentation.
- Constructed and deployed an automated data-governance dashboard on Azure Databricks using Delta Live Tables materialized views, exposing catalog, schema, table privileges, user activity, and pipeline run history from Unity Catalog system tables, deployed as a Databricks Asset Bundle operating under a service principal.
- Led a feasibility study on Databricks-to-Azure SQL connectivity across five methods (Lakehouse Federation, Spark JDBC, Unity Catalog JDBC, server-side DML, PyTDS) on both Serverless and Classical compute, validating read and write paths over private endpoints and VNet injection, and subsequently defined the production adoption architecture.
- Established on-premises data gateway integrations to transfer SQL Server tables and flat files into a Microsoft Fabric Lakehouse, and installed the OneStream SIC gateway with Simba Spark ODBC and OAuth service-principal authentication, enabling direct querying of Databricks from the finance platform.

- Executed ITIL-based on-call monitoring across 6 Databricks workspaces and 2 platform environments with a 4-hour proactive sweep cadence and a 15-minute P1 SLA, supporting a multi-hop Delta Share pipeline feeding an on-premises SCADA system, managing incidents in ServiceNow and PagerDuty, and authoring post-incident Standard Operating Procedures (SOPs).
- Migrated multi-year legacy SPSS survey data to Azure on a medallion architecture using Synapse Analytics and Data Factory, incorporating event-driven schema-change detection across 10+ survey types and role-based controls, reducing manual processing by approximately 70%.
- Developed cargo-operations pipelines and a data-governance framework on Azure Logic Apps with integrated validation rules and pipeline-health monitoring, and rebuilt a payroll data platform on Data Factory and Synapse with automated validation compliant with data-privacy requirements.

University of Ottawa, Ottawa, Canada | Research Assistant & Teaching Assistant 2020 – 2023

- Developed approximate Bayesian neural networks for inference-time uncertainty quantification in probabilistic classification, enhancing reliability for clinical decision support, with findings published in IEEE.
- Constructed a multimodal emotion and bipolar-severity assessment system using datasets exceeding 2 million frames, integrating VGG-Face and FER2013 visual features, a deep belief network for audio, early fusion techniques, and an LSTM for video classification.
- Instructed tutorials for groups of up to 180 students on topics including uncertainty evaluation in machine learning, computer architecture, and electrical circuits.

Tamtaam Development Company | Backend & Database Developer 2016 – 2019

- Designed and optimized MS SQL Server databases, including normalization, indexing, and stored procedures, and automated data loading processes using SSIS ETL packages.
- Developed C#/ASP.NET backends, configured REST APIs, and integrated third-party services.

EDUCATION

- University of Ottawa**, M.Sc., Electrical & Computer Engineering (Software Engineering), Tehran, Iran 2020 – 2022
- Shahid Beheshti University**, B.Sc., Electrical & Computer Engineering (Power Systems), Tehran, Iran 2011 – 2015

SKILLS

Microsoft Fabric

Azure

Databricks

ETL/ELT pipeline design

Medallion architecture

Delta Lake

Dimensional modeling

Star schema

Data warehousing

SSIS migration

Unity Catalog

Azure DevOps CI/CD

ARM templates

On-premises data gateway

OneStream SIC

ODBC/JDBC

REST APIs

Python

SQL

Spark SQL

T-SQL

C#

ASP.NET

PyTDS

sqlglot

Streamlit

Git

RAG

LangGraph

Microsoft Presidio

RoBERTa

PyTorch

TensorFlow

scikit-learn

HuggingFace Transformers

Client-facing delivery

Architecture documentation

Stakeholder presentation

ITIL incident management

SOP authoring

Root cause analysis

Agentic AI workflows

Hybrid dense+sparse retrieval

Qdrant

LLM evaluation (synthetic golden datasets)

Human-in-the-loop workflows

Azure AI Foundry

Azure OpenAI

Anthropic Claude

Azure ML managed endpoints

Kubernetes

NER / PII de-identification

Bayesian neural networks

Uncertainty quantification

LSTM

Multimodal learning

PySpark

Azure Data Factory

Synapse Analytics

Azure Logic Apps

Delta Live Tables

Databricks Asset Bundles

Lakehouse Federation

Delta Share

LANGUAGES

English (C1), Persian (native), French (A1)

CERTIFICATES AND CLEARANCES

Databricks Certified Generative AI Engineer Associate, Databricks

Government of Canada reliability status : security clearance

09/2026